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KASANA LUWEERO SECONDARY SCHOOLS ASSOCIATION
(KALUSSA) JOINT MOCK EXAMINATIONS

535/1

PHYSICS

Paper 1

July, 2026

Uganda Certificate of Education

***Paper* 1**

Theory

2 *hours* 30 *minutes*

INSTRUCTION TO CANDIDATES:

*This paper consists of **two** sections; **A** and **B** it has seven examination items.*

*Section **A** has **three** compulsory items.*

*section **B** has **two** parts; **I** and **II**. Answer **one** question from **each** part.*

*Answer **five** items in all.*

Any additional item(s) answered will not be scored.

*All answers **must** be written in the booklets provided.*

SECTION A

(*All items in this section are compulsory)*

1. During the daytime journey while traveling for a study trip, some learners asked the driver why part of the windscreen was covered with stickers. The driver explained that some passengers get frightened by an illusion on the road that looks like a pool of water, especially on hot days though they couldn't understand how it occurs.

   At night, the bus interior lights were $cyan$, and they noticed that their dresses, which appeared $magenta$ in daylight, changed color under the cyan lighting. This surprised them, and they debated the reason for the change.

   Later, the learners tried tuning to certain radio station from Kampala, but as the bus moved toward Jinja, the signal became weak and eventually disappeared. They wondered how radio waves behave and if other electromagnetic waves can as well be useful in everyday life.

   Hint:

   - $Radio\ station\ broadcasts\ at\ a\ frequency\ of\ 100\ MHz$
   - $Speed\ of\ electromagnetic\ waves\ =\ 3.0\ x10^{8}\ ms^{-1}$
   - $A\ technician\ told\ them\ that\ the\ wave\ length\ of\ radio\ signals\ that\ reach\ Jinja\ from\ Kampala\ should\ not\ exceed\ 10\ m.$

   Task: You have been chosen to discuss the observations made during the trip.

   a) The appearance of the leaners dress in the bus lights in the night.

   b) The phenomena that were based on to put stickers on the wind screen of the bus that the driver tried explaining to the learners.

   c) If the wave length of the radio signals could reach their location

   d) An application of the concept that brings about the appearance of a pool of water.

2. Whenever the parents of a patient under taking radiotherapy from $Cobalt - 60$ want to check on him in the isolated room, the doctor tells them to wait for the right time.

The doctor has also informed the relatives that if the right time for them to check on the patient comes, they must observe precautions in the isolated room.

The relatives have demanded to know the period they should wait before they can check on their patient but no one answers them.

They looked up on the hospital ceiling, there was a smoke detector, the doctor said that the device uses radioisotope called Americium 241.

The parents were left wondering how such a dangerous substance can be used in daily life, what precautions are always taken, and why they need to wait for that time.

You have heard the doctor talking to the relatives and they have contacted you to address their concerns.

$Activity\ (counts\ per\ minute)$	480	280	170	100	60	35
$Time\ (hours)$	0	10	20	30	40	50

Table 1

($Hint$: $The\ right\ time\ is\ when\ the\ activity\ is\ equal\ to\ the\ background\ count\ rate\ of\ 40$

$counts\ per\ minute$)

Task: As learner of Physics help the parents of the patient;

a) Know how long they should wait before they are allowed to check on their patient. (Use graphical method to analyze the data)

b) Understand why they need to wait for that time before they enter the isolated room.

c) Know the precautions the parents have to observe inside the isolated room.

d) Understand how the radioactivity material is applied in smoke detector.

3. In a certain country, a T.V reporter was reporting live at night near the ocean about the high tides. Viewers in another country were watching the live broadcast of the news bulletin during day time instead. The viewers wondered how it could be day and night at the same time, and how the event in one country could be watched live on TV in another country. They also questioned how the moon brings about low and high tides.

Task: Using your knowledge of physics, help the viewers to understand their concerns.

PART 1

*Attempt **one** item from this part.*

4. After driving for 8 $minutes$ at a constant speed on a downhill highway, a fully loaded cargo truck driver suddenly notices a big tree fallen on the road. The driver applies the brakes and the truck takes 80 $seconds$ to come to rest, slightly colliding with the fallen tree. During the incident, some goods were moved forward due to sudden stop. Authorities arrive and observe that:

- *The truck was heavily loaded, tyres were worn out and a large amount of heat was produced during braking*
- *The truck traveled a total distance of* 5.3 km *from when it started constant speed to when it uniformly decelerated to rest*
- *If the truck used steel brakes of mass* 25 kg *per wheel and* 7000 kJ *of heat was generated at each wheel.*

Hint:

- $The\ insurance\ company\ compensates\ only\ if\ the\ driver\ was\ within\ the\ required\ speed\ limits\ of\ 40\ kmh^{-1}$
- $Specific\ heat\ capacity\ of\ steel\ brakes\ =\ 470\ J\ kg^{-1}K^{-1},$
- $initial\ temperature\ =\ 20°C$
- $Maximum\ efficient\ brake\ temperature\ range\ =\ 600\ °C\ to\ 700\ °C$

Task: Help the truck driver to;

(a) Determine whether the company will compensate for the damages.

(b) Understand reasons why the loaded truck took long to stop and why the there was large heat during braking.

(c) determine whether the brakes were suitable for use

d) Explain the application of friction in breaking the car.

5. Omega cycles daily to school, which is about $10\ km$ from his home. One morning, he set off at exactly $7{:}00\ a.m.,$ carrying hot coffee in a metal bottle. As he rode along, he noticed that his bicycle was producing a continuous sound whenever he pedaled. The noise made his journey uncomfortable, and he wondered what could be causing it.

After travelling about $6\ km$, Omega checked the time and found it was already $7{:}40\ a.m.$, meaning he was likely to be late. To save time, he increased his speed. His velocity increased from about $2.5\ m\ s^{-1}\ to\ 6.0\ m\ s^{-1}$ over a short period of $5\ seconds$, which helped him arrive at school on time. Later, his classmates asked him about his acceleration, but he could not work it out.

During lunch break, Omega opened his bottle expecting to find his coffee still hot, but it was only warm. Disappointed, he shared this with a friend, who advised him to use a vacuum flask because it can keep liquids hot for a longer time. He became curious about how the vacuum flask works to reduce heat loss and where else the method of heat losses is applied in everyday life.

Task: Omega has approached you for assistance to;

(a) Explain the likely cause of the noise and heat while riding and suggest how it can be minimized.

(b) Determine his acceleration during the journey.

(c) Explain how a vacuum flask keeps liquids hot

(d) Understand any application of the process that led to the cooling of the coffee.

SECTION B

PART II

*Attempt **one** item from this part.*

6. During a science exhibition to demonstrate how electricity is generated, transmitted, and distributed to consumers, one group of learners presented a model showing the journey of electrical energy from a power station to homes and small businesses.

Their teacher visited the exhibition and was impressed by the learners' work.

The school management is concerned about the electricity bills of some staff member who uses the following electrical appliances in her home (*see list below*) and complains that her monthly electricity bill of $UGX\ 200{,}000$ is higher than expected.

> $Four\ security\ light\ bulbs\ rated\ 240\ V, 5\ W, used\ for\ 10\ hours\ per\ night.$
>
> $Electric\ kettle\ rated\ 240\ V, 2000\ W, used\ once\ daily\ for\ 5\ hours.$
>
> $Cost\ of\ a\ unit\ of\ electricity\ is\ 650/=$
>
> $A\ budget\ of\ 200{,}000/=\ \ is\ available\ for\ monthly\ electricity\ bills$

Task: You have been requested to help him with the following concerns;

a) Explain how electricity reaches consumers from the power station.

b) Justify the type of connections recommended for domestic appliances

c) Determining whether her electricity bill is reasonable

d) Explain the application of the transformer used in running low voltage domestic appliances.

7. During a heavy thunderstorm, a few minutes later, fire broke out inside the cinema building. After the fire was extinguished, investigations revealed that a television set connected to an old rooftop antenna caused the fire. The antenna had not been properly grounded, and a technician suspect that lightning struck the antenna, sending a high flow of current through the cable into the TV.

He advised that an automatic circuit breaker should be installed in addition to a device that arrests lightning and the right fuse. However, the manager of the cinema hall could not understand these instructions.

Hint;

- *The TV is rated 340 W 220 V*
- *The technician said that if a fuse of resistance 1.6 A was installed in the TV, large current would not have not destroyed the TV.*

Task: Help the cinema hall manager understand;

(a) how the device recommended protects the house from the incidence.

(b) know the resistance of the fuse to be used.

(c) how the recommended circuit breaker protects appliances.

END